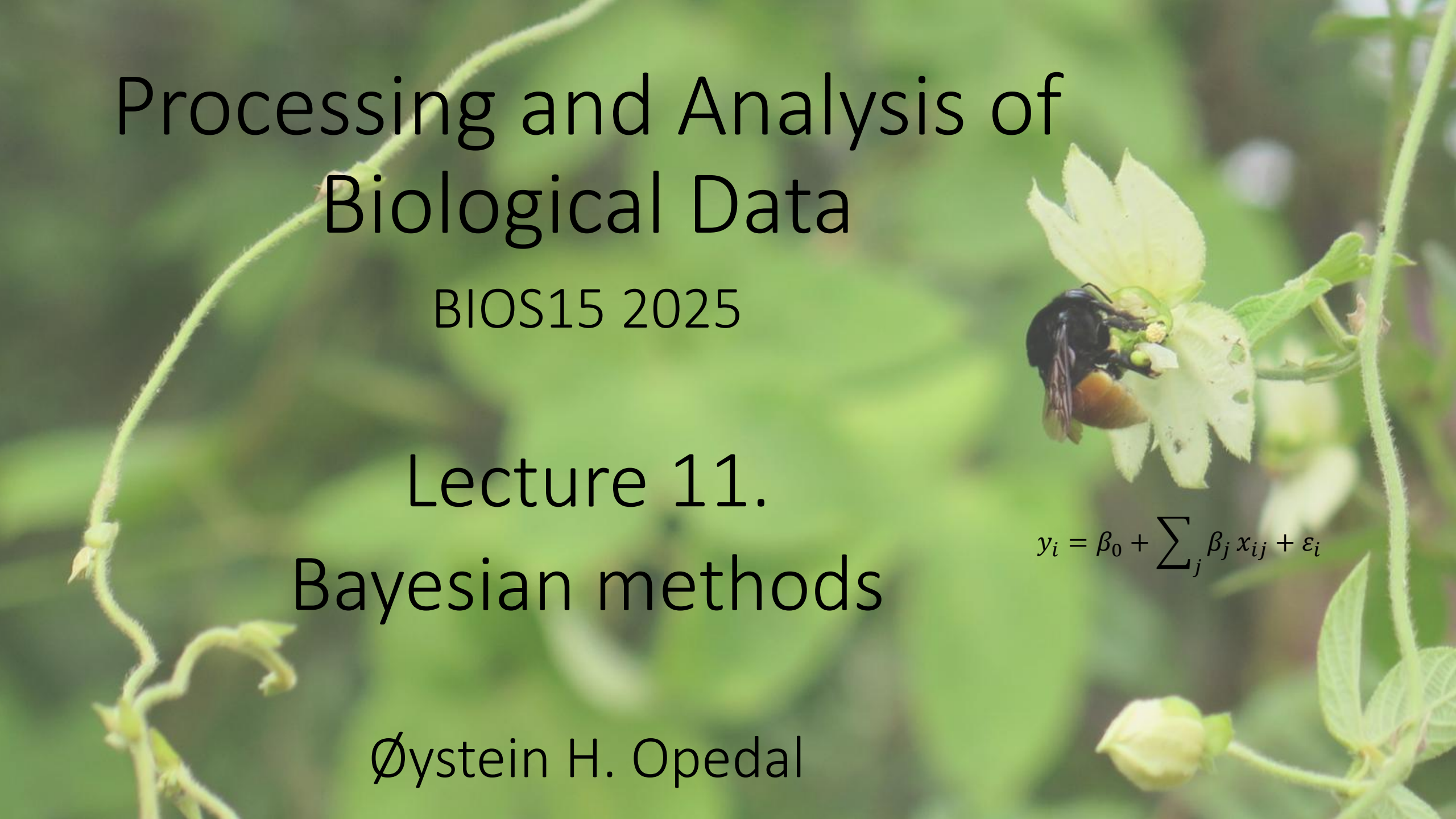
The last week!

- Today:
  - Lecture on **Bayesian methods**
  - Seminar on **the *Viscaria* project** (Sophie & Aarushi)
  - Exercise session (Bayesian methods)
- Wednesday
  - Summary day
  - General comments and example report (GLM)
  - Exam advice
  - Question hour
  - No exercise (?)

Processing and Analysis of Biological Data

BIOS15 2025



$$y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i$$

Lecture 11. Bayesian methods

Øystein H. Opedal

Evaluating support without P-values

- Interpret parameter estimates and their uncertainty (SE, 95% CI)
- AIC model selection (e.g. comparing complex model to null model)
- Related: Causal inference through SEM – support for overall model
- Bayesian inference (today's topic)
- Use language to reflect support:
  - Both species were more commonly observed in warmer microhabitats (Table 1)
  - Two out of three species tended to be observed more frequently in warmer microhabitats, although with variable statistical support (Table 1)
  - Never use "significant" alone ("statistically significant" if anything)

Frequentist vs. Bayesian statistics

- There are two main schools or approaches to statistical analysis: **frequentist** approaches and **Bayesian** approaches
- So far we have fitted models with least-square or maximum likelihood methods. These are frequentist methods yielding a test statistic we can compare to a known distribution
- In practise, biologists often use a mix of approaches, depending on the task or availability of software/R packages





Rev. T. Bayes

Bayesian methods

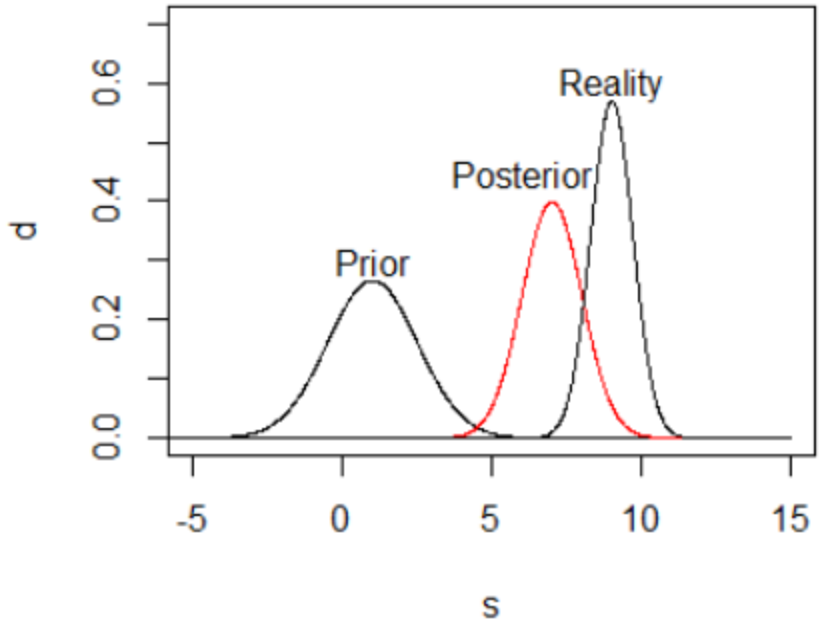
- Bayesian methods differ from frequentist approaches by incorporating information about our prior belief in different posible outcomes
- The prior distribution is defined before model fitting, which yields the posterior distribution
- (Sometimes) more focus on the distribution of posible values than on the most likely value

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(\theta|S) = \frac{P(S|\theta)P(\theta)}{\int P(S|\theta)P(\theta)d\theta}$$

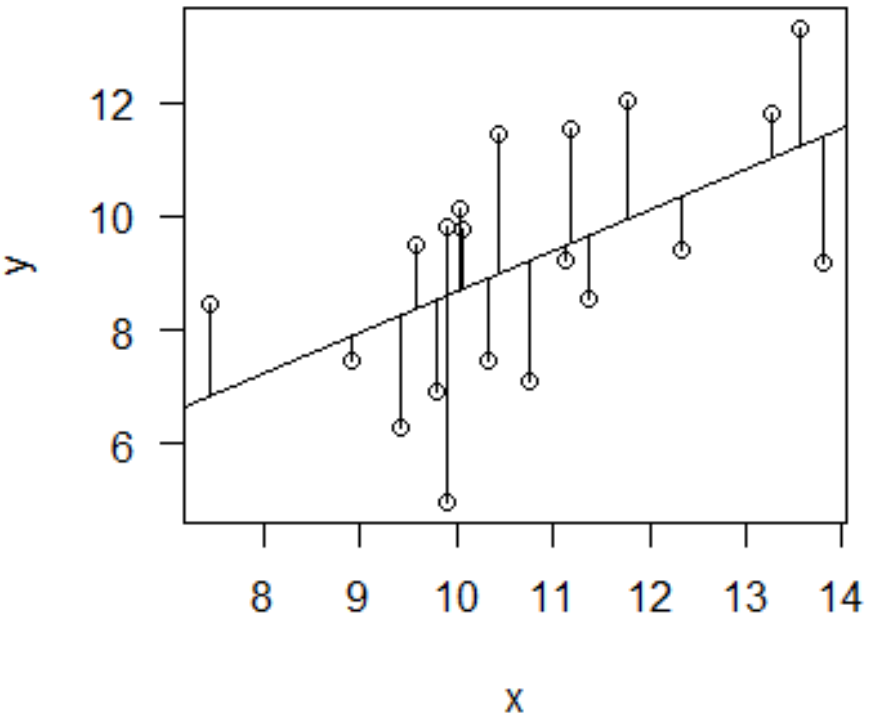
Prior and posterior distributions

- The prior distribution reflects our belief in different values of a given parameter
- After model fitting (posterior sampling), the posterior distribution will (hopefully!) be closed to reality than is the prior



The linear model

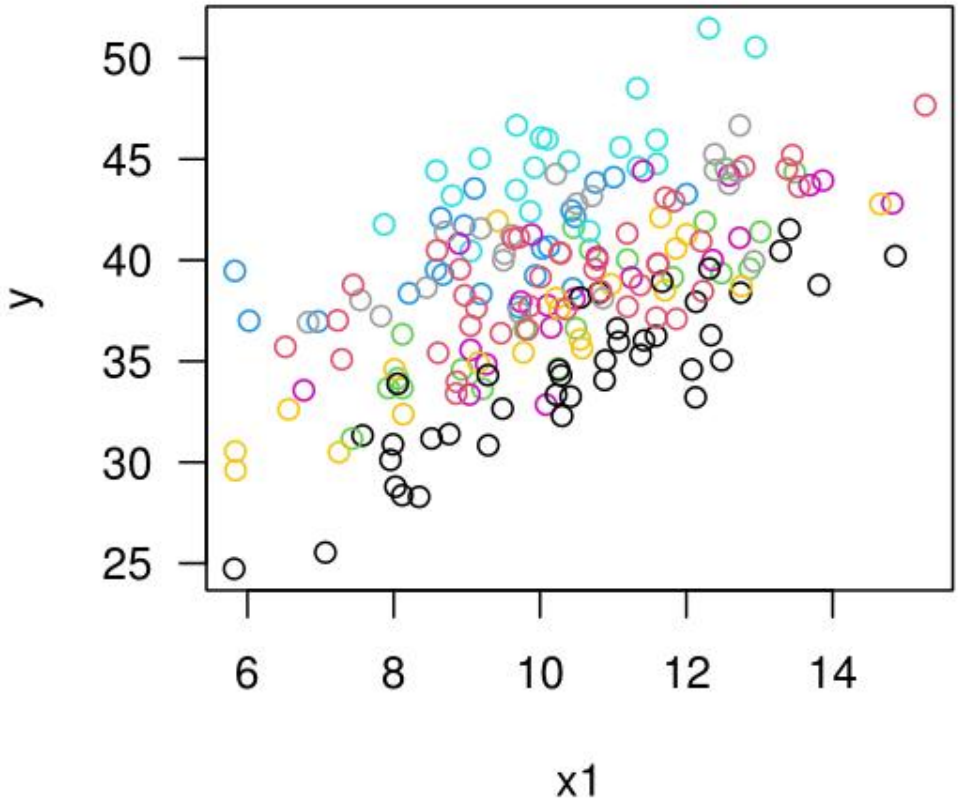
- Most of the models we will work with in this course are linear models, that describe how a linear set of predictors relate to a response variable
- A key element of the model is the so-called linear predictor:
- $y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i, \varepsilon \sim N(0, \sigma^2)$
- The term $\varepsilon \sim N(0, \sigma^2)$ means that the residuals (epsilon) are assumed to follow a normal distribution



Linear mixed models

$$y = \mathbf{X}\beta + \mathbf{Z}\mu + \epsilon$$

- Linear mixed models extend the linear model by introducing **random effects**, i.e. variables for which we are more interested in the variance than the mean
- Often used to account for hierarchical structure of sampling (e.g. plots nested in forest patches)
- Assumes normally distributed group means, and normally distributed residuals within groups

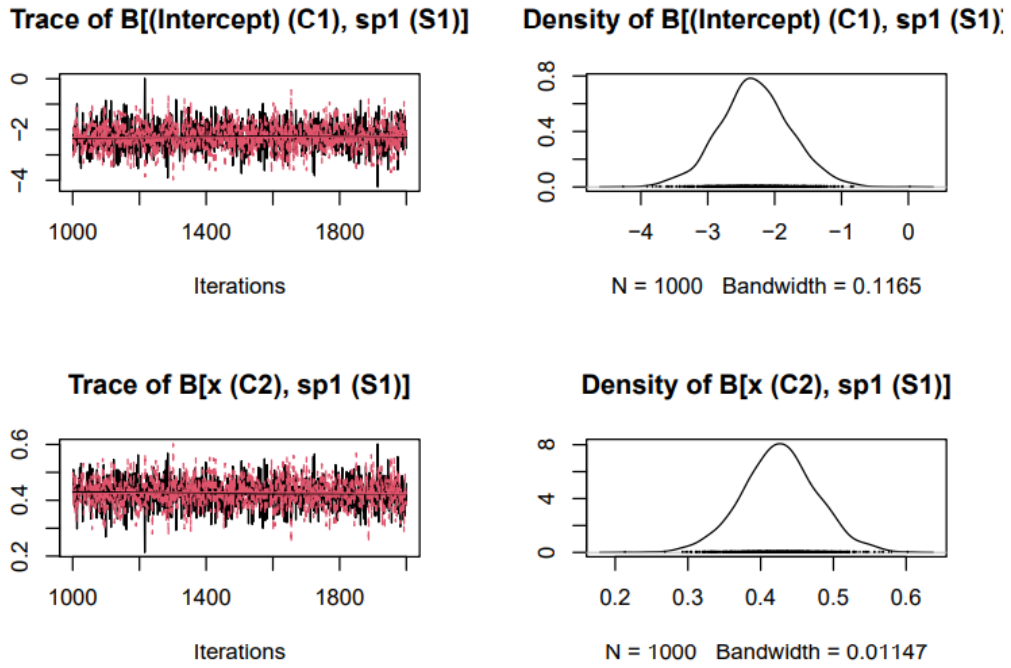


Posterior sampling (“model fitting”)

- The posterior distribution is sampled by an algorithm, typically the Markov chain Monte Carlo (MCMC)
- Briefly, the algorithm evaluates the likelihood of the data under a series of parameter sets, each a small modification of the previous solution, taking into account the prior belief in different values of the parameter.
- Over time the chain approaches the “best” available solution

Evaluating model convergence

- After MCMC posterior sampling, we have to evaluate that the sampler has reached a stable posterior distribution



Evaluating model convergence

- We can evaluate mixing and convergence quantitatively by computing **effective sample sizes** (posterior simples accounting for autocorrelation)
- With multiple replicate chains, we can also assess how well the results correspond by **potential scale reduction factors** (psrf)

```
effectiveSize(mpost$Beta)
```

```
## B[(Intercept) (C1), sp1 (S1)]          B[x (C2), sp1 (S1)]
##                       1850.331                      1858.394
```

```
gelman.diag(mpost$Beta, multivariate=F)$psrf
```

```
##                                Point est. Upper C.I.
## B[(Intercept) (C1), sp1 (S1)]   1.001291   1.003076
## B[x (C2), sp1 (S1)]             1.001872   1.004578
```

Fitting a linear model with MCMC

- Several packages, e.g. `MCMCglmm`
- Here we will use `Hmsc`, because we will also use this package for fitting joint models
- In mixed models, the variance component parameter is Omega.

```
##
## Iterations = 1001:2000
## Thinning interval = 1
## Number of chains = 2
## Sample size per chain = 1000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##                                      Mean      SD Naive SE Time-series SE
## B[(Intercept) (C1), sp1 (S1)] -2.2933 0.53332  0.01193       0.012414
## B[x (C2), sp1 (S1)]            0.4263 0.05144  0.00115       0.001194
##
## 2. Quantiles for each variable:
##
##                                     2.5%     25%     50%     75%   97.5%
## B[(Intercept) (C1), sp1 (S1)] -3.3825 -2.6344 -2.3035 -1.9612 -1.2245
## B[x (C2), sp1 (S1)]            0.3235  0.3928  0.4257  0.4592  0.5302
```

```
getPostEstimate(m2, "Omega")$mean
```

```
##          [,1]
## [1,] 11.02465
```

Joint species distribution modelling

A fast emerging field in statistical community ecology

2017

ECOLOGY LETTERS

Ecology Letters, (2017) **20**: 561–576 doi: 10.1111/ele.12757

IDEA AND PERSPECTIVE

How to make more out of community data? A conceptual framework and its implementation as models and software

Otso Ovaskainen,[1,2]* Gleb Tikhonov,[1] Anna Norberg,[1] F. Guillaume Blanchet,[3,4] Leo Duan,[5] David Dunson,[5] Tomas Roslin[6] and Nerea Abrego[2,7]

Abstract

Community ecology aims to understand what factors determine the assembly and dynamics of species assemblages at different spatiotemporal scales. To facilitate the integration between conceptual and statistical approaches in community ecology, we propose Hierarchical Modelling of Species Communities (HMSC) as a general, flexible framework for modern analysis of community data. While non-manipulative data allow for only correlative and not causal inference, this framework facilitates the formulation of data-driven hypotheses regarding the processes that structure

2020

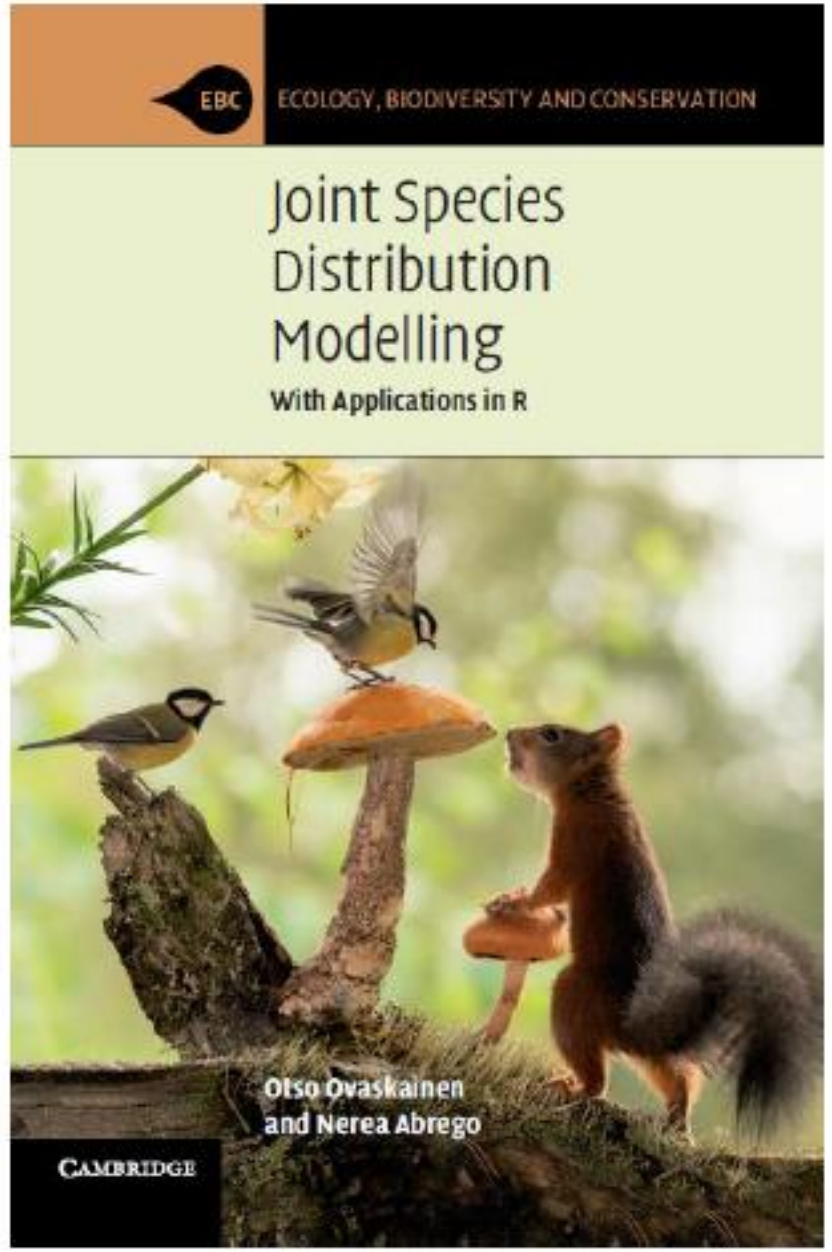
APPLICATION

Methods in Ecology and Evolution

Joint species distribution modelling with the R-package HMSC

Gleb Tikhonov[1,2] | Øystein H. Opedal[2,3] | Nerea Abrego[4] | Aleksi Lehikoinen[5] | Melinda M. J. de Jonge[6] | Jari Oksanen[7] | Otso Ovaskainen[2,3]

2020



Example:
birds in Swedish areas of high conservation value

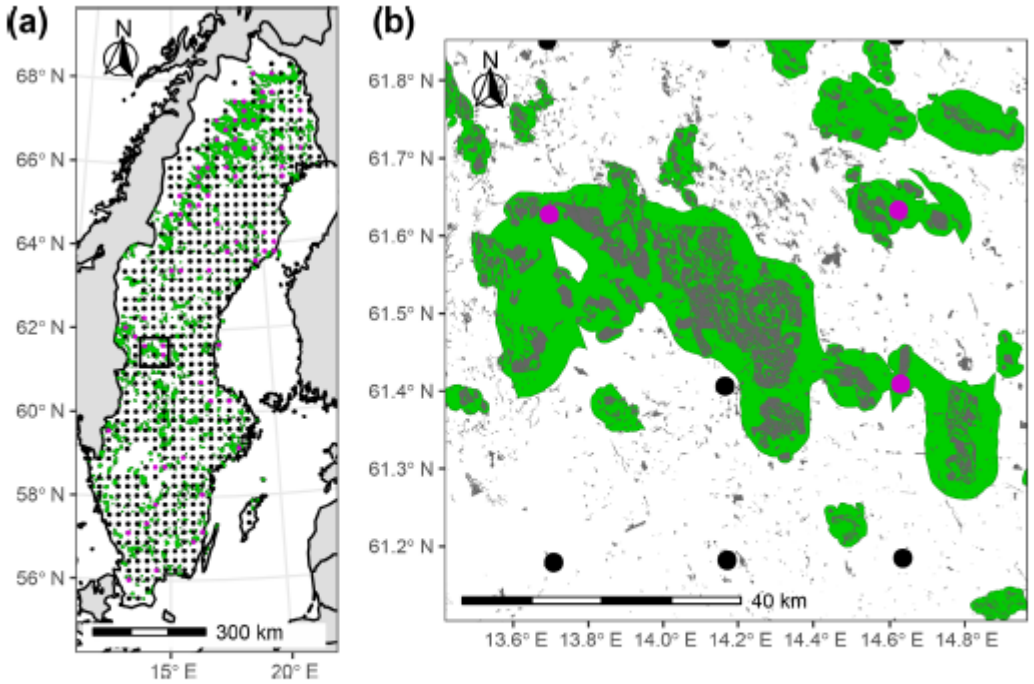
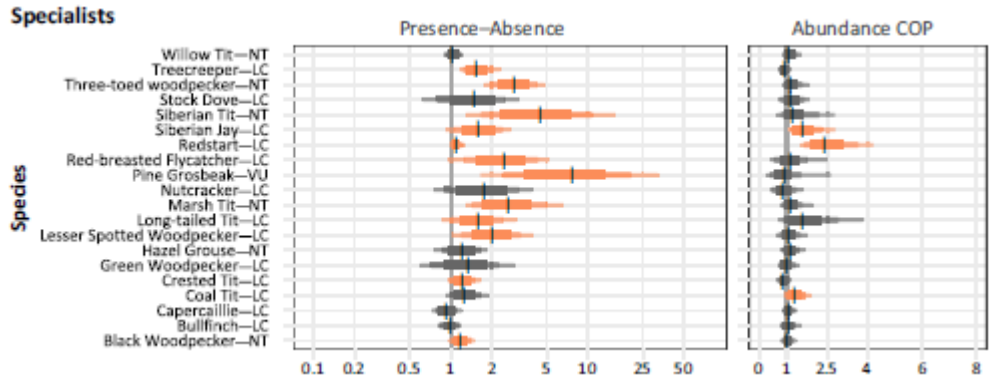


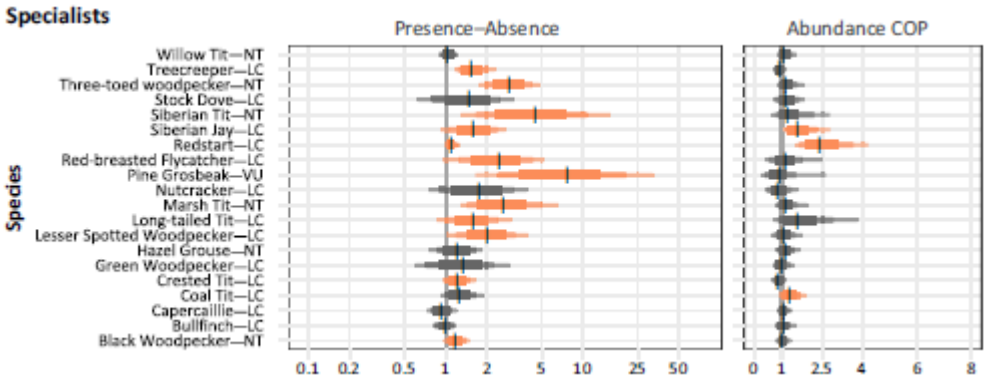
FIGURE 1 (a) Map of the proposed areas of high conservation value (AHCVs, in green) defined by Bovin et al. (2017), with the outline of map (b) shown as a black square. (b) Example of individual AHCVs (in green) with their constituent valuable forests (protected forests as well as unprotected forests that are classified as valuable for nature conservation) in gray. The AHCVs are landscapes with at least 1000 ha of forest that have a high density of valuable forests. Black dots for the centroid of routes outside AHCVs and pink dots for the centroid of routes inside AHCVs.



Bakx et al. 2023 Ecosphere

Example:
birds in Swedish areas of high conservation value

- By modelling the distribution of many species jointly, we can “borrow” information across species (e.g. use data on common species to train models for rare species)
- We can also use patterns of species associations to make predictions, as in “indicator species”



Bakx et al. 2023 Ecosphere

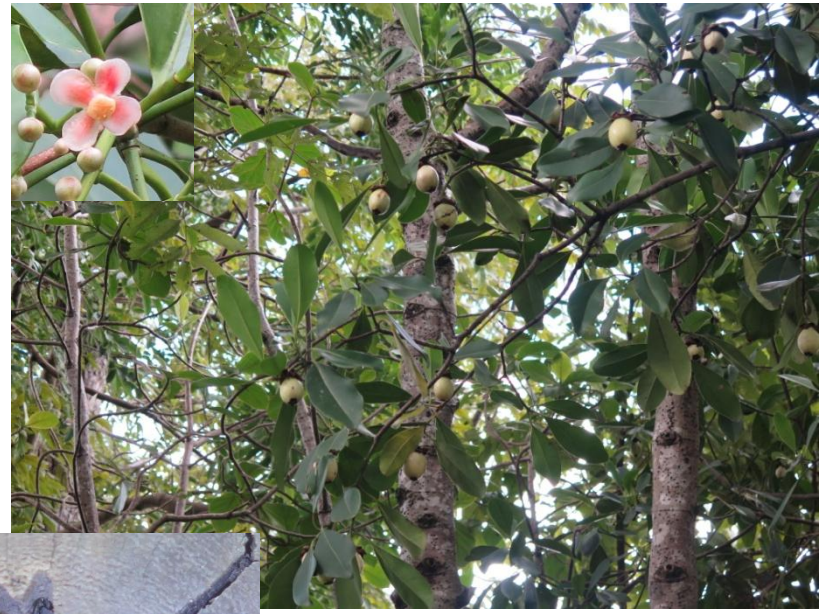
Euglossine bees

- Tribe Euglossini (Apidae)
- *Eulaema*, *Eufriesea* & *Euglossa*
- *Exaerete* & *Aglae* (cleptoparasites)
- ~230 species, all occur in tropical America. Typically one forth of bee species in tropical forests
- Mostly solitary, sometimes colonial and primitively eusocial



Female resource needs

- Nectar
- Pollen
- Resin for nest construction



Clusia sp.
(Clusiaceae)



Euglossa dilemma
Dalechampia scandens



Eufriesea sp.
Clusia sp.



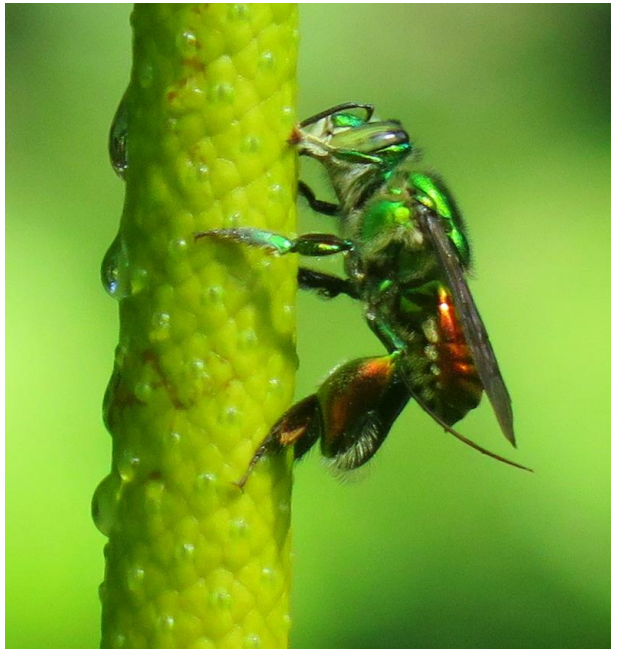
Photos by E. Albertsen, S. Jaeger, Ø. Opedal

Male resource needs

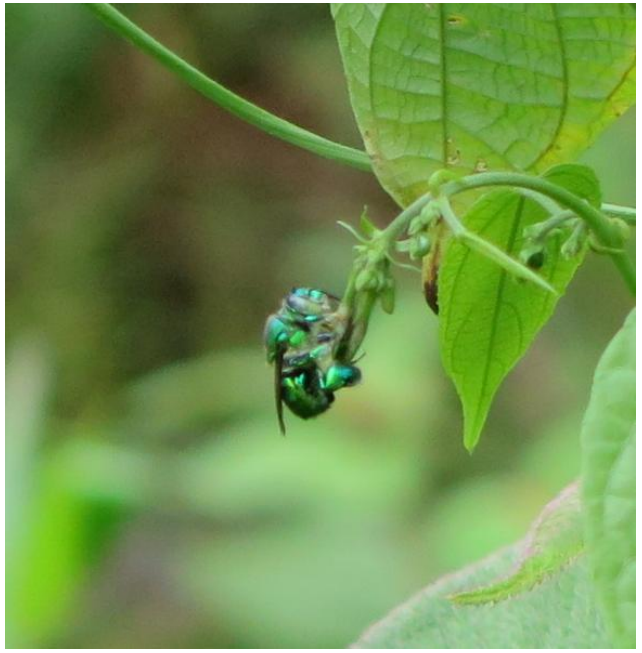
- Nectar
- Fragrances for courtship displays



Euglossa hansoni on *Gongora*
Photo by S. Ramírez



Euglossa cf. *ignita* on *Anthurium*



Euglossa sp. on *Dalechampia*

Attracting male euglossine bees to fragrance baits



Euglossa dilemma



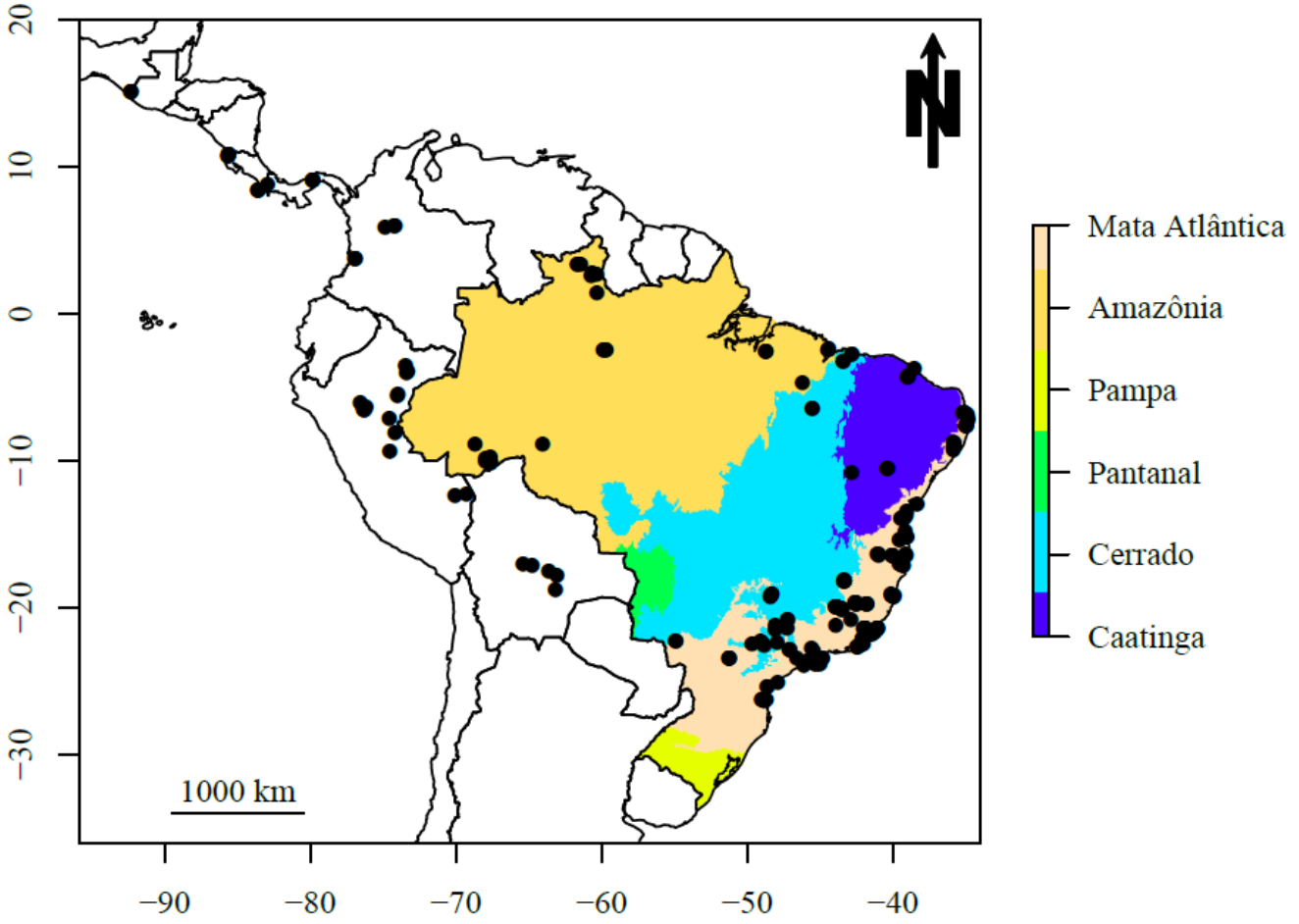
Eufriesea pulchra



Eulaema meriana

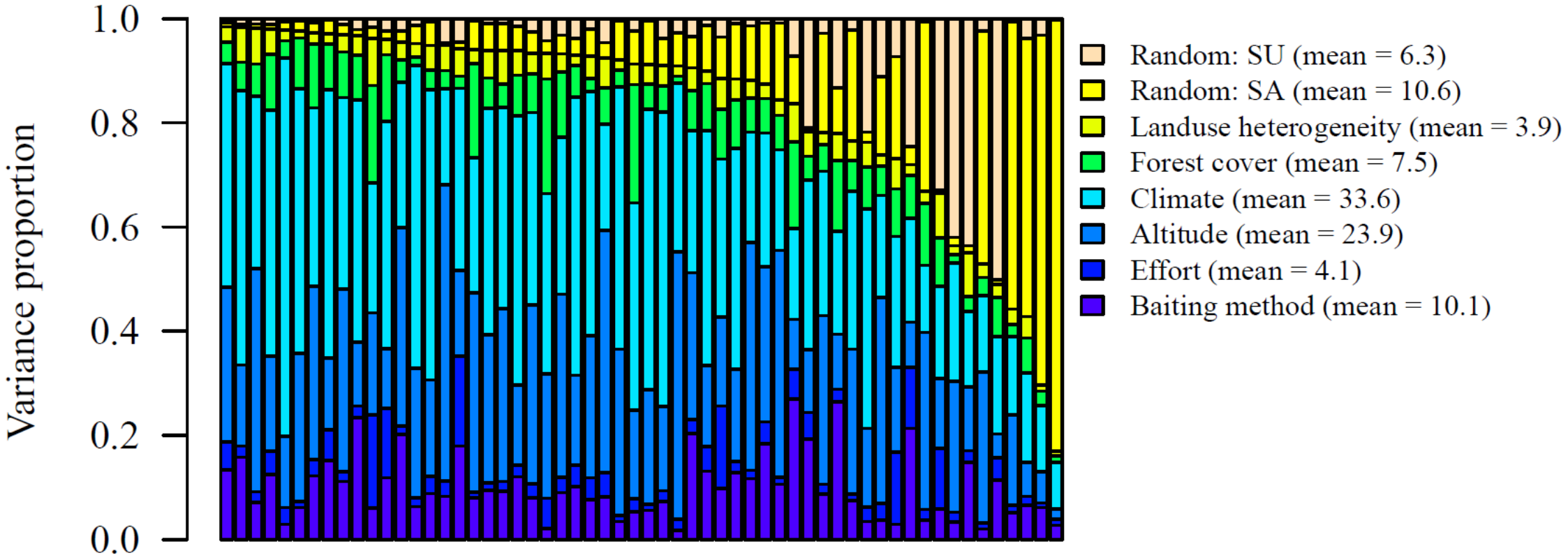
A database and synthesis of euglossine bee assemblages at fragrance baits

- 173 species, 129 study sites, 297 samples
- Data on bait use
- Some trait data, including bait preferences
- Available in GitHub: github.com/oysteiop/eugcomm

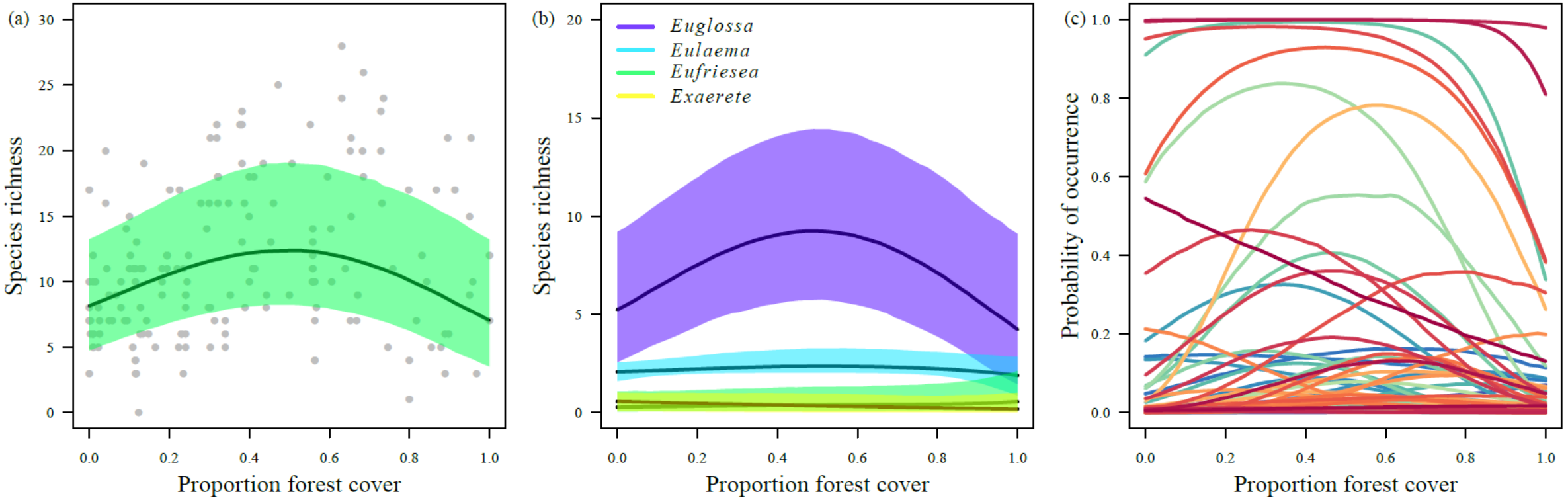


Opedal, Martins & Marjakangas 2020 Apidologie

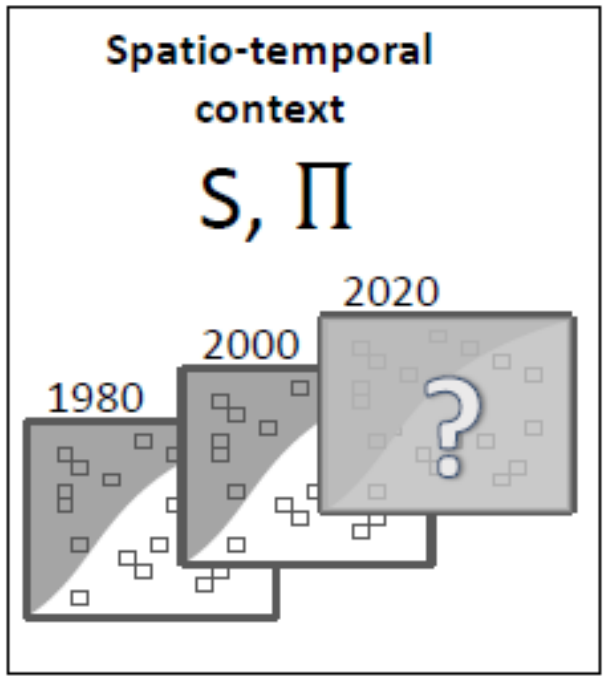
Effects of climate and landscape structure on Atlantic forest euglossines

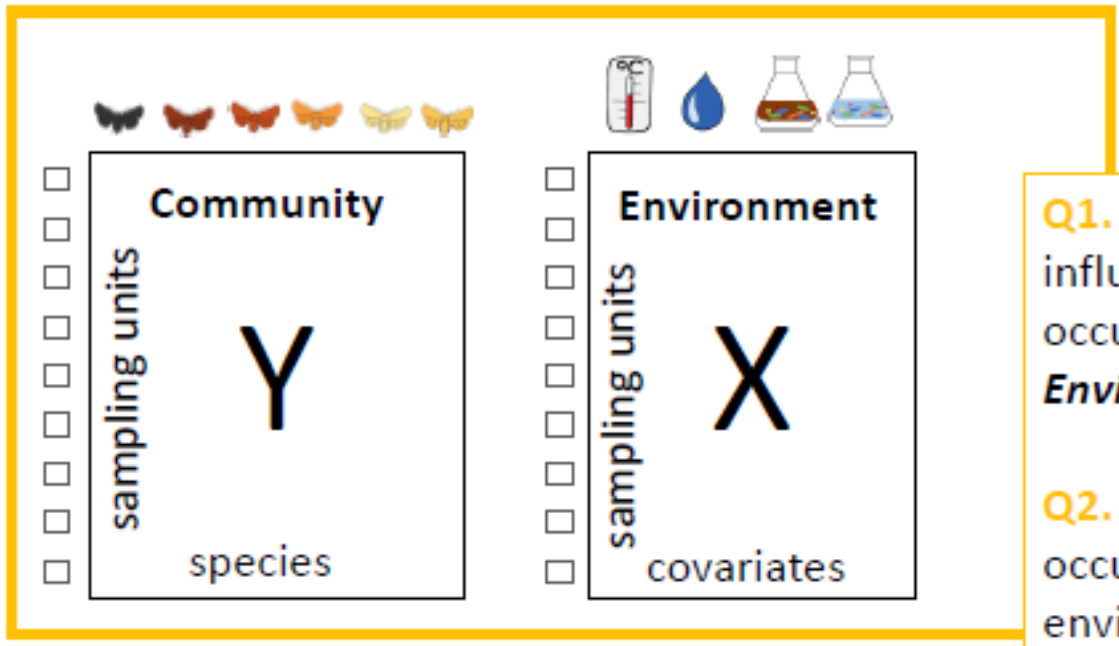


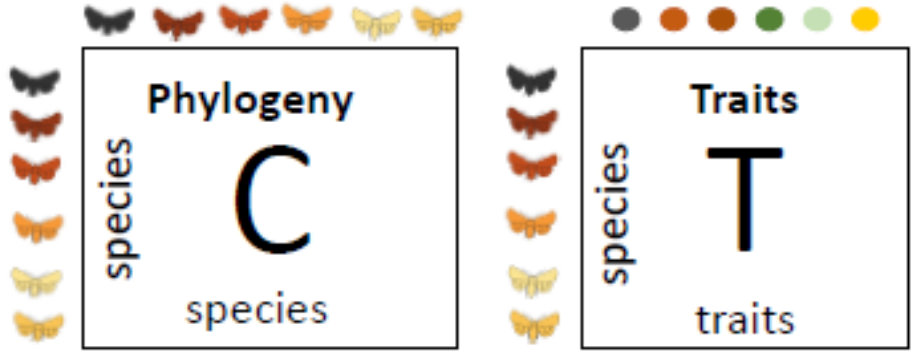
Effects of climate and landscape structure on Atlantic forest euglossines



Typical questions asked in community ecology





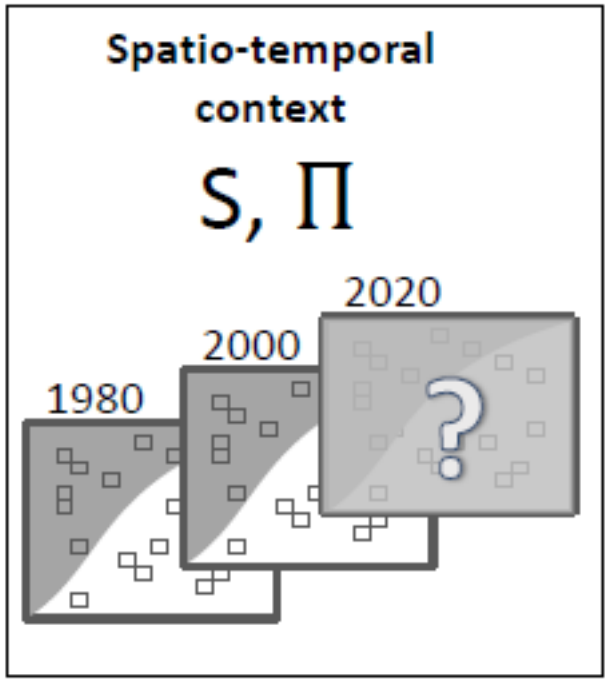


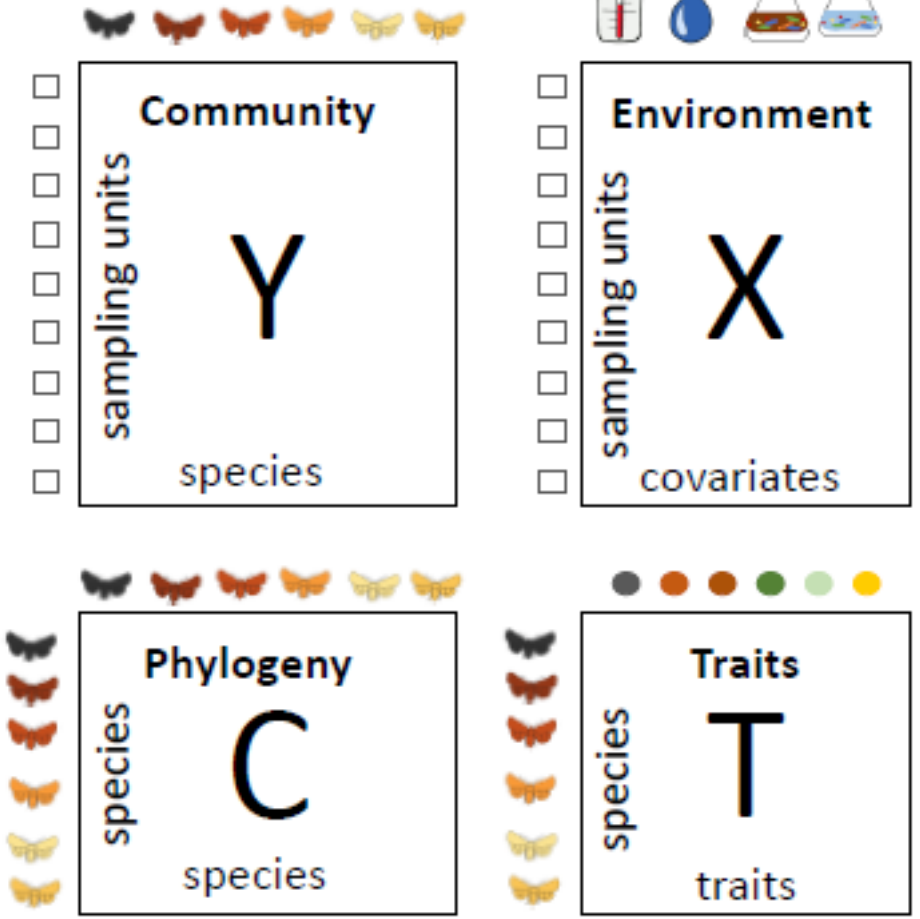
Q1. How do environmental conditions influence species occurrences/community composition? ***Environmental filtering***

Q2. What are the species' co-occurrence patterns after the effects of environmental conditions are taken into account? ***Biotic filtering***

Typical questions asked in community ecology

Q5. How will communities change over space and time? ***Prediction***





Q1. How do environmental conditions influence species occurrences/community composition? ***Environmental filtering***

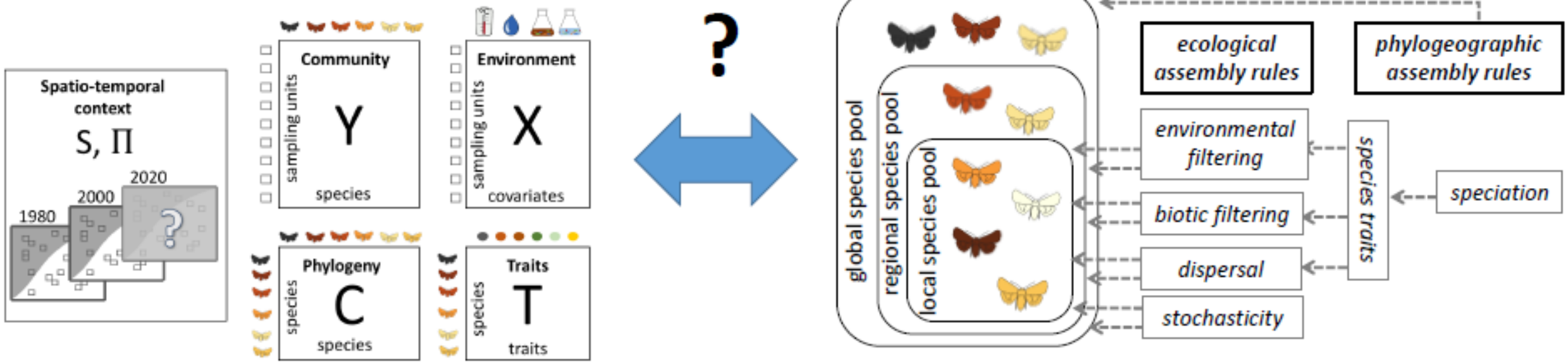
Q2. What are the species' co-occurrence patterns after the effects of environmental conditions are taken into account? ***Biotic filtering***

Q3. How do traits influence the responses of the species to the environment?

Q4. Are the responses of the species to the environment phylogenetically structured? ***Phylogeographic processes***

The BIG QUESTION in *statistical* community ecology:

How to best use the data to address the research questions?



Overview of (typical) methods in community ecology: which data they utilize?

Method	Input data (as a whole, without subsetting or pre-treatment)				
	Community data	Environmental data	Trait data	Phylogenetic data	Spatio-temporal structure
Ordinations	✓	~	~	~	~
Co-occurrence analyses	✓	✗	✗	✗	✗
Network analyses	✓	✗	✗	✗	✗
Diversity metrics	✓	~	✗	✗	~
Joint species distribution modelling	✓	✓	✓	✓	✓

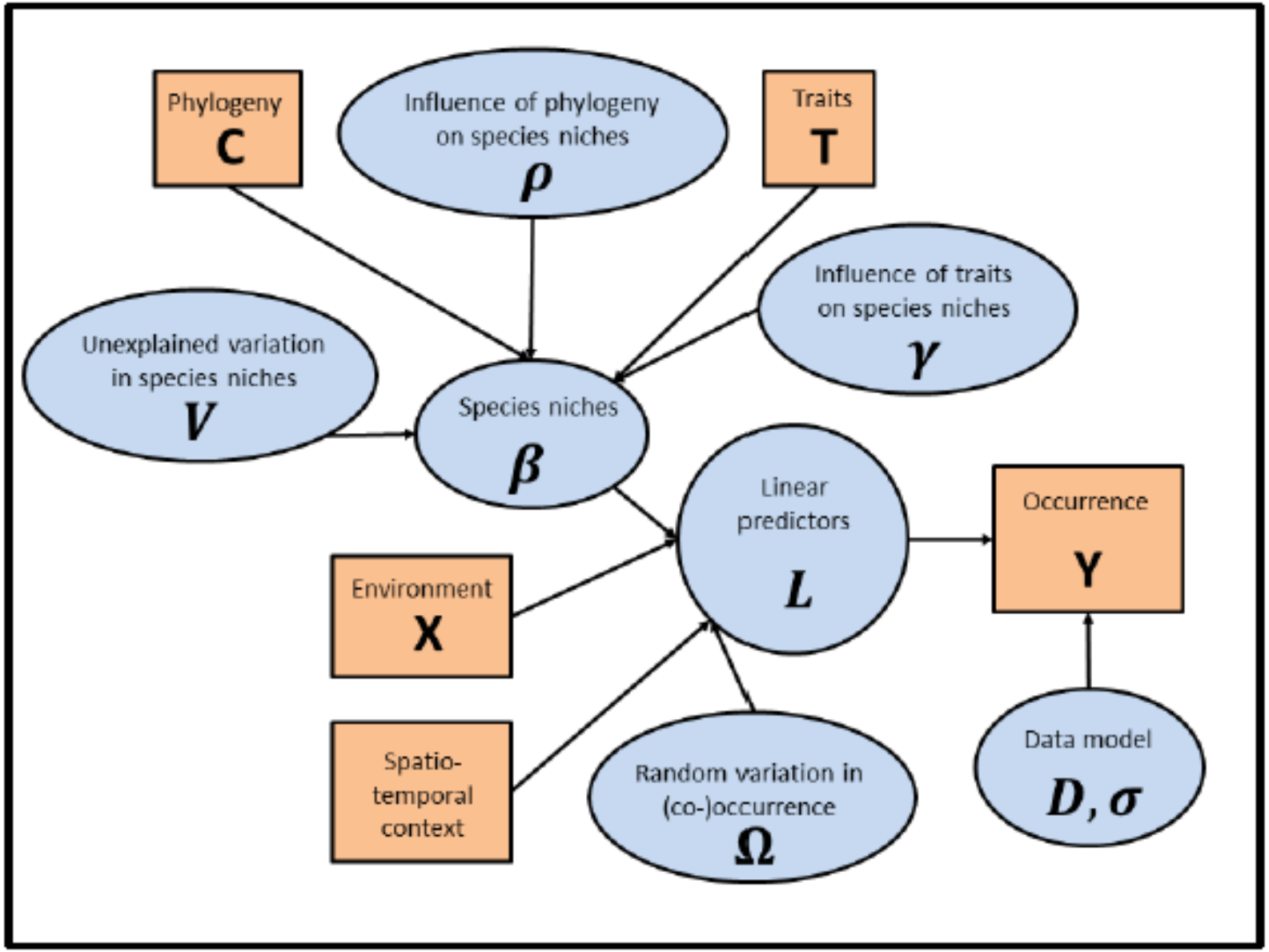
✓Yes, ~Some variants, ✗No

Overview of (typical) methods in community ecology: which kinds of questions they answer?

	Outputs (without post-hoc treatment)			
Method	**Environmental filtering**	**Biotic filtering**	**Effects of traits and phylogeny**	**Spatiotemporal predictions**
Ordinations	✓	~	~	✗
Co-occurrence analyses	✗	✓	✗	✗
Network analyses	✓	✓	✗	✗
Diversity metrics	✓	✗	✗	~
Joint species distribution modelling	✓	✓	✓	✓

✓Yes, ~Some variants, ✗No

Hierarchical Modelling of Species Communities (HMSC)



Graphical description of the HMSC model as a Directed Acyclic Graph (DAG)

HMSC is a multivariate hierarchical generalized linear mixed model fitted with Bayesian inference

Fixed effects Random effects

Linear predictor for sampling unit i and species j: $$L_{ij} = L_{ij}^F + L_{ij}^R$$

Examples of data models:

Linear model: $$y_{ij} = L_{ij} + \varepsilon_{ij}$$

Probit model: $$y_{ij} \sim \text{Bernoulli}(\Phi(L_{ij}))$$

Lognormal Poisson model: $$y_{ij} \sim \text{Poisson}\left(\exp\left(L_{ij} + \varepsilon_{ij}\right)\right)$$